

Augmented Programs — NodeJS Express To-do + React Quiz

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Design a to-do list application using NodeJS and ExpressJS.
- Design a Quiz app using ReactJS.
- MongoDB University certification is included as a separate guide.

2. Workflow

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Workflow: input → processing → data/state → output

3.1. Code pathway — index.html

```
1 <!doctype html><html><head><meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1"><title>FSD2 React Quiz</title></head><body><div id="root"></div><script type="module" src="/src/main.jsx"></script></body></html>
```

3.2. Code pathway — main.jsx

```
1 import React, { useState } from 'react';
2 import { createRoot } from 'react-dom/client';
3
4 const questions = [
5   { q: 'Which hook stores state?', options: ['useState', 'useRoute', 'useHTML'], answer: 'useState' },
6   { q: 'Which method is used to find MongoDB documents?', options: ['find()', 'show()', 'open()'], answer: 'find()' },
7   { q: 'Express is mainly used for?', options: ['Server-side web apps', 'Image editing', 'Operating systems'], answer: 'Server-side web apps' }
8 ];
9
10 function App() {
```

4. Runtime walkthrough

Runtime walkthrough — Augmented Programs — NodeJS Express To-do + Re



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?